Chapter 16, Problem 72P

Problem

The transfer function of a system is

$$H(s) = \frac{s^2}{3s + 1}$$

Find the output when the system has an input of $14e-t/3u(t)$.

Step-by-step solution

Step 1 of 4

The input to the system is,

$$x(t) = 4e^{-\frac{t}{3}}u(t)$$

Apply Laplace transforms on both sides.

$$X(s) = \mathcal{L}\left\{4e^{-\frac{t}{3}}u(t)\right\}$$
$$= \frac{4}{s + \frac{1}{3}}$$

Calculate the output of the system by using the transfer function of the system.

$$H(s) = \frac{s^2}{3s+1}$$
$$\frac{Y(s)}{X(s)} = \frac{s^2}{3s+1}$$

$$Y(s) = \frac{s^2}{3s+1}X(s)$$
$$= \frac{s^2}{3s+1} \times \frac{4}{s + \frac{1}{3}}$$
$$= \frac{12s^2}{(3s+1)^2}$$
$$= \frac{12s^2}{9s^2 + 6s + 1}$$

Comments (3)

Anonymous

its fucking 14 in the book bitches

Anonymous

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Anonymous

it should be 42s^2

Step 2 of 4

Apply long division method.

$$\begin{array}{rl} 9s^2+6s+1\big) & 12s^2 \qquad \left(\frac{4}{3}\right. \\ - & \underline{12s^2+8s+\frac{4}{3}} \\ & \qquad -8s-\frac{4}{3} \end{array}$$

The modified expression for $Y(s)$ is,

$$\begin{aligned} Y(s) &= \frac{4}{3} - \frac{8s+\frac{4}{3}}{(3s+1)^2} \\ &= \frac{4}{3} - \frac{8s+\frac{4}{3}}{(3s+1)^2} \end{aligned}$$

Step 3 of 4

Apply partial fractions to some part of the expression.

$$\frac{8s+\frac{4}{3}}{(3s+1)^2} = \frac{A}{(3s+1)^2} + \frac{B}{3s+1}$$

$$8s+\frac{4}{3} = A + B(3s+1)$$

$$8s+\frac{4}{3} = (3B)s + (A+B)$$

Compare s terms on both sides.

$$3B = 8$$

$$B = \frac{8}{3}$$

Compare constant terms on both sides.

$$A+B = \frac{4}{3}$$

$$A+\frac{8}{3} = \frac{4}{3}$$

$$A = -\frac{4}{3}$$

Step 4 of 4

The modified expression for $Y(s)$ is,

$$Y(s)=\frac{4}{3}-\frac{8}{3}\left(\frac{1}{3s+1}\right)+\frac{4}{3}\left[\frac{1}{(3s+1)^2}\right]$$
$$=\frac{4}{3}-\frac{8}{9}\left(\frac{1}{s+\frac{1}{3}}\right)+\frac{4}{27}\left[\frac{1}{\left(s+\frac{1}{3}\right)^2}\right]$$

Apply inverse Laplace transforms.

$$y(t)=\frac{4}{3}\delta(t)-\frac{8}{9}e^{-\frac{t}{3}}u(t)+\frac{4}{27}te^{-\frac{t}{3}}u(t)$$
$$=\frac{4}{3}\delta(t)-\frac{4}{9}\left(2-\frac{t}{3}\right)e^{-\frac{t}{3}}u(t)$$

Thus, the output of the system is $\boxed{\frac{4}{3}\delta(t)-\frac{4}{9}\left(2-\frac{t}{3}\right)e^{-\frac{t}{3}}u(t)}$.

Comments (1)

Anonymous

Instead of 4/3 and 4/9, it's 14/3 and 14/9.